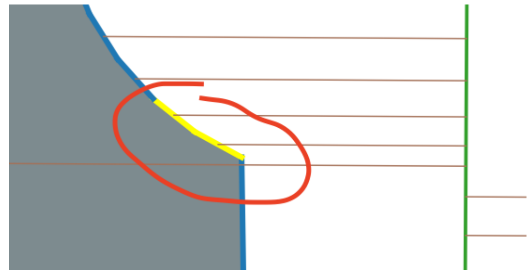
Improving Street Representation in GIS

Benefits:

- In the lot line labelling algorithm, this would fix the problem of oblique fronts being identified as sides, as is common in cul-de-sacs.

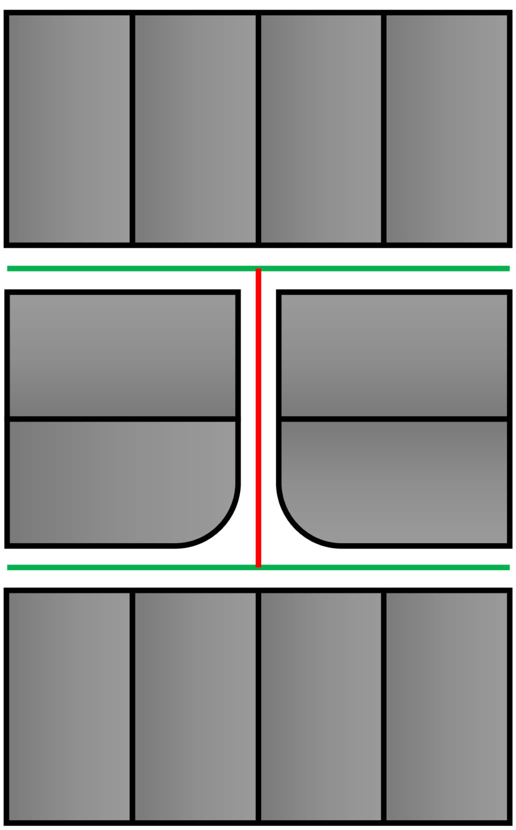


Steps:

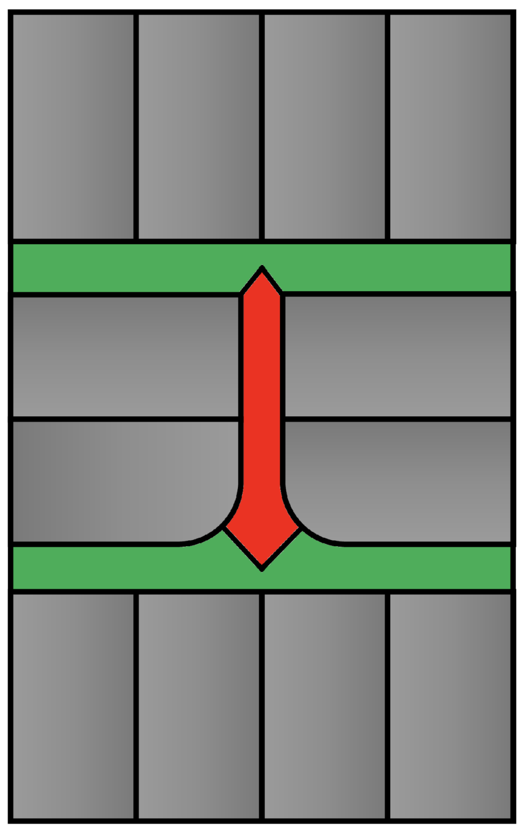
1. For each point not in a parcel, determine the closest street or lane.
2. Group points by closest street or lane and generate a polygon to represent each.
3. Trim 0.01 m from each street and lane polygon to ensure there is a small gap along parcels, as is needed for the lot line algorithm to generate bee-lines.

In the diagrams below parcels are grey, streets are green, and lanes are red.

Before:



After:



Challenges:

- It is hard to handle streets on the edges of municipalities. There might need to be a maximum distance threshold when determining which points to include in each polygon.